COMPOSITE MATERIAL DATASHEETS CFRP G0926/RTM6-5HS

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MATERIAL IDENTIFICATION

Short informal name	G0926/RTM6-5HS
Commercial name	Hexcel HexForce G0926 D1304 TCT INJ E01 2F (reinforcement fabric) Hexcel HexFlow RTM6 (resin)
Specifications	Airbus: AIMS05-04-009E (reinforcement fabric) AIMS05-04-100 (resin)
Description	Carbon Fibre Reinforced Polymer (CFRP) fabric produced by resin infusion
Version	0.1 (01/04/2022)

REFERENCES

- NT-F-ID-00051 Is. 4 "QUALIFICATION TEST REPORT FOR RTM6/G0926D 1304 TCT INJ E01 2F (Standard cure cycle)" Airbus Test Report
- QTP130004 Is. 1 "Qualification of RTM6/G0926 D 1304 TCT INJ E01 2F: additional tests at 120ºC" Airbus Qualification Test Program
- QTR130004 Is. 2 "Qualification of RTM6/G0926 D 1304 TCT INJ E01 2F (standard cure cycle): additional tests" Airbus Qualification Test Report
- RP0612716 Is. 3 "Material allowables of standard modulus carbon fibre fabric, 5H Satin, 370 g/m2 with liquid epoxy resin 180 ºC curing as per AIMS 05-04-009 (Grade C and E)" Airbus Technical Report
- ME-DN-NT-120028 Is. C "Analysis of Test Results in coupons of Hexcel G0926/RTM6 fabric laminates" Airbus DS Technical Report
- TR4957 Is. 1 "Interim report - G0926D Qualification - HT application" Hexcel Report
- ESCRIBIR NT DEDICADA CON COMALL 0.21 ETC…

PHYSICAL PROPERTIES

Cured Ply Thickness (*CPT*)	0.363	mm
Density	1510	kg/m³
Tg-onset (wet)	173	ºC
Max. Operational Limit (*MOL*)	120	ºC

Coefficients of Thermal Expansion CTE (1/ºC)	Range	-55…20ºC	20…70ºC	70…120ºC
	α_1	3.4E-06	3.4E-06	2.2E-06
	α_2	3.4E-06	3.4E-06	2.2E-06
	α_3	4.9E-05	5.6E-05	7.2E-05

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -	K_B -
Values	64300	64300	4700	0.08	0.795

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13} -	ν_{23} -
Values	17400	5200	5200	0.371	0.371

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1, E_2 and G_{12} in buckling stability analyses

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis	E-KDF			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{1t}	861.3	817.4	801.9	781.7	0.9	0.949	0.931	0.908	775.2	735.7	721.7	703.5
F_{1c}	676.7	521.7	471.5	377	0.696	0.771	0.697	0.557	471	363.1	328.2	262.4
F_{2t}	861.3	817.4	801.9	781.7	0.9	0.949	0.931	0.908	775.2	735.7	721.7	703.5
F_{2c}	676.7	521.7	471.5	377	0.696	0.771	0.697	0.557	471	363.1	328.2	262.4
F_{12}	98.8	82.4	74.7	49.9	0.609	0.834	0.756	0.505	60.1	50.2	45.5	30.4

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values (με)				B-basis	E-KDF			B-basis Values (με)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
ε_{tall}	13400	12700	12500	12100	0.897	0.948	0.933	0.903	12000	11400	11200	10900
ε_{call}	10500	8100	7300	5900	0.696	0.771	0.695	0.562	7300	5600	5100	4100

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis	E-KDF			B-basis Values			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{13} (MPa)	66.4	51.2	44	30.9	0.488	0.772	0.664	0.465	32.4	25	21.5	15.1
F_{3t} (MPa)												
G_{IC} (N/mm)												
G_{IIC} (N/mm)												

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis	E-KDF			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
OHT_{QI}	356.8	356.8	356.8	356.8	0.9	1	1	1	321.1	321.1	321.1	321.1
OHC_{QI}	313	246.3	225.3	175.3	0.871	0.787	0.72	0.56	272.8	214.7	196.4	152.8
FHT_{QI}	360	360	360	360	0.9	1	1	1	324	324	324	324
FHC_{QI}	406.6	378.3	-	266.3	0.577	0.93	-	0.655	234.4	218.1	-	153.6
F_{BR-QI}	599.9	538.7	-	495.7	0.787	0.898	-	0.826	472.2	424	-	390.2
F_{PO}												

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

Property	B-basis Value for all conditions
TAI (με)	

$E/t^2 = A \cdot (1 - e^{-B \cdot d0/t})$

Property	Design Value
A (J/mm²)	
B	

Approximation model type: SMM chapter 1421: CDTM-T05

$CAI = K_B \cdot EKDF \cdot \{ CAI_{\infty\text{-}QI} \cdot (3 / K_{TOH})^{Cx} + A / e^{Be \cdot (E/t^{\wedge}2)} \}$

Property	Mean Value RT/Dry	B-basis K_B	*E-KDF*		
			70ºC/W	90ºC/W	120ºC/W
$CAI_{\infty\text{-}QI}$ (με)					
A (με)					
B_e					
C_x					

…where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

$$K_{TOH} = 1 + \{ 2 \cdot [(E_x/E_y)^{0.5} - \nu_{xy}] + E_x/G_{xy} \}^{0.5}$$

(*) CAI shall be calculated not only for the 0º direction (X parallel to material 0º reference), but also for the rest of ply orientations, i.e. the 45º and 90º directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x, E_y, G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the alowable strains applicable to plies at 0º (CAI_0), at ±45º (CAI_{45}) and at 90º (CAI_{90}).